WORKED SOLUTION

STEP II 1999, Question 8

Pure • Analysis

STEP II 1999, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s2-q8

The key idea

Multiplying $\sin k\theta$ by $2\sin\frac{1}{2}\theta$ turns it, by a product-to-sum formula, into a *difference* of two cosines, and the sum then telescopes. After that, both parts are about choosing θ well: with $\theta = \pi/n$ the awkward cosine $\cos(n+\frac{1}{2})\theta$ becomes a simple multiple of $\cos\frac{1}{2}\theta$. For (ii), the half-angle formula turns $\sin^2$ into a cosine, and the sum $\sum k\cos k\theta$ is exactly what differentiating $(*)$ produces.

Solution

Proving $(*)$

Assume $\sin\frac{1}{2}\theta \neq 0$, so that the right-hand side makes sense. The identity $2\sin A\sin B = \cos(A-B) - \cos(A+B)$ with $A = k\theta$ and $B = \frac{1}{2}\theta$ gives

$$2\sin\tfrac{1}{2}\theta\,\sin k\theta = \cos\left(k-\tfrac{1}{2}\right)\theta - \cos\left(k+\tfrac{1}{2}\right)\theta.$$

Summing from $k=0$ to n, each $\cos(k+\frac{1}{2})\theta$ subtracted in row k is added back in row $k+1$, so only the first and last terms survive:

$$2\sin\tfrac{1}{2}\theta\sum_{k=0}^{n}\sin k\theta = \cos\left(-\tfrac{1}{2}\theta\right) - \cos\left(n+\tfrac{1}{2}\right)\theta = \cos\tfrac{1}{2}\theta - \cos\left(n+\tfrac{1}{2}\right)\theta,$$

since cosine is even. Dividing by $2\sin\frac{1}{2}\theta$ gives $(*)$.

Part (i)

Put $\theta = \pi/n$ (with $n \geq 1$), so $\sin\frac{1}{2}\theta = \sin\frac{\pi}{2n} \neq 0$. Then

$$\cos\left(n+\tfrac{1}{2}\right)\theta = \cos\left(\pi+\tfrac{\pi}{2n}\right) = -\cos\tfrac{\pi}{2n},$$

so the numerator of $(*)$ is $2\cos\frac{\pi}{2n}$ and

$$\sum_{k=0}^{n}\sin\left(\frac{k\pi}{n}\right)=\frac{2\cos\frac{\pi}{2n}}{2\sin\frac{\pi}{2n}}=\cot\frac{\pi}{2n}.$$

This is exact. When n is large, $\frac{\pi}{2n}$ is small, so $\cos\frac{\pi}{2n}\approx 1$ and $\sin\frac{\pi}{2n}\approx\frac{\pi}{2n}$, giving

$$\sum_{k=0}^{n}\sin\left(\frac{k\pi}{n}\right)\approx\frac{1}{\pi/(2n)}=\frac{2n}{\pi}.$$

Part (ii)

By the double-angle formula, $\sin^2\frac{k\pi}{2n}=\frac{1}{2}\left(1-\cos\frac{k\pi}{n}\right)$, so with $\theta=\pi/n$

$$\sum_{k=0}^{n}k\sin^2\left(\frac{k\pi}{2n}\right)=\frac{1}{2}\sum_{k=0}^{n}k-\frac{1}{2}\sum_{k=0}^{n}k\cos k\theta=\frac{n(n+1)}{4}-\frac{1}{2}\sum_{k=0}^{n}k\cos k\theta. \qquad (1)$$

To find $\sum k\cos k\theta$ we differentiate $(*)$ with respect to θ; the left-hand side becomes exactly this sum. First split the right-hand side as

$$\frac{1}{2}\cot\tfrac{1}{2}\theta-\frac{\cos(n+\frac{1}{2})\theta}{2\sin\frac{1}{2}\theta}.$$

Write $S=\sin\frac{1}{2}\theta$ and $c=\cos\frac{1}{2}\theta$. The first piece differentiates to $-\frac{1}{4}\operatorname{cosec}^2\frac{1}{2}\theta=-\frac{1}{4S^2}$. By the quotient rule the second piece differentiates to

$$\frac{(n+\frac{1}{2})\sin(n+\frac{1}{2})\theta\cdot 2S+\cos(n+\frac{1}{2})\theta\cdot c}{4S^2}.$$

So

$$\sum_{k=0}^{n}k\cos k\theta=\frac{-1+(2n+1)S\sin(n+\frac{1}{2})\theta+c\cos(n+\frac{1}{2})\theta}{4S^2}.$$

Ask yourself Why put $\theta=\pi/n$ *after* differentiating, not before?

The identity $(*)$ holds for all θ (with $\sin\frac{1}{2}\theta\neq 0$), so we may differentiate it. If we fixed $\theta=\pi/n$ first we would have a single number with nothing to differentiate. Once we have the derivative as an identity in θ, we can substitute any value we like.

Now take $\theta=\pi/n$. As in (i), $(n+\frac{1}{2})\theta=\pi+\frac{1}{2}\theta$, so $\sin(n+\frac{1}{2})\theta=-S$ and $\cos(n+\frac{1}{2})\theta=-c$. Hence

$$\sum_{k=0}^{n}k\cos\frac{k\pi}{n}=\frac{-1-(2n+1)S^2-c^2}{4S^2}=\frac{-2-2nS^2}{4S^2}=-\frac{1}{2S^2}-\frac{n}{2},$$

using $c^2=1-S^2$. Substituting into (1):

$$\sum_{k=0}^{n}k\sin^2\left(\frac{k\pi}{2n}\right)=\frac{n(n+1)}{4}+\frac{n}{4}+\frac{1}{4S^2}=\frac{n(n+2)}{4}+\frac{1}{4\sin^2\frac{\pi}{2n}}.$$

This is exact. For large n, $\sin\frac{\pi}{2n} \approx \frac{\pi}{2n}$, so $\frac{1}{4\sin^2(\pi/2n)} \approx \frac{n^2}{\pi^2}$, and $\frac{n(n+2)}{4} = \frac{n^2}{4} + \frac{n}{2}$, whose second term is negligible beside n^2. Therefore

$$\sum_{k=0}^{n} k\sin^2\left(\frac{k\pi}{2n}\right) \approx \left(\frac{1}{4} + \frac{1}{\pi^2}\right) n^2,$$

as required.

Checks

- $n = 1$ in (ii): the sum is $0 + 1\cdot\sin^2\frac{\pi}{2} = 1$, and the exact formula gives $\frac{3}{4} + \frac{1}{4} = 1$.
- $n = 2$ in (i): the sum is $0 + 1 + 0 = 1$, and $\cot\frac{\pi}{4} = 1$.
- Numerically, for $n = 1000$ the sum in (i) is $636.6192\ldots$ against $2n/\pi = 636.6198\ldots$, and the sum in (ii) is 351 821.3, within 0.15% of $(\frac{1}{4} + \frac{1}{\pi^2})n^2$. The relative error in (ii) shrinks like $1/n$ because of the neglected $\frac{n}{2}$ term.

Common slips

- Writing $\cos(\pi + x) = \cos x$: the correct value is $-\cos x$, and that sign is what doubles the numerator in (i).
- Forgetting the $\frac{1}{2}\sum k = \frac{n(n+1)}{4}$ term in (ii), which supplies the $\frac{1}{4}n^2$.

Beyond STEP

Part (i) is a Riemann sum in disguise: $\frac{\pi}{n}\sum\sin\frac{k\pi}{n} \to \int_0^{\pi}\sin x\,\mathrm{d}x = 2$, which is exactly the statement $\sum \approx 2n/\pi$. Similarly, (ii) says $\frac{1}{n^2}\sum k\sin^2\frac{k\pi}{2n} \to \int_0^1 x\sin^2\frac{\pi x}{2}\,\mathrm{d}x = \frac{1}{4} + \frac{1}{\pi^2}$. The same telescoping trick applied to $\sum\cos k\theta$ gives the Dirichlet kernel of Fourier analysis. Abbott's *Understanding Analysis* treats Riemann sums and Fourier series carefully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.